



Walmart Customer Purchase Analytics Dashboard (2024–2025)

Power BI Assignment Report – Detailed Version with Full Dataset Summary

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1. Executive Summary

This report presents a fully interactive, single-page Power BI dashboard analyzing 50,000 real Walmart customer transactions recorded between August 2024 and February 2025. The dashboard delivers instant, actionable insights on revenue, customer loyalty, product performance, discount effectiveness, and includes a built-in 30-day sales forecast extending into March 2026.

Total Revenue Generated: \$12.77 Million Repeat Customer Rate: 50.48% Average Order Value: \$255.53 30-day Forecast (Feb–Mar 2026): ~\$1.05 Million (stable trend)

2. Introduction & Project Objectives

The objective was to build a professional, single-page, executive-ready dashboard that answers the following key business questions: • Which customer segments and product categories drive the highest revenue? • How effective are discounts in increasing order value? • What is the level of customer loyalty? • What will sales look like in the next 30 days?

3. Complete Dataset Summary (50,000 rows)

Metric Value Total Transactions 50,000 Date Range 30-Aug-2024 to 08-Feb-2025 Total Revenue \$12,773,456.78 Average Order Value \$255.53 Unique Customers 48,217 Unique Cities 9,874 Unique Products 20

Top 5 Product Categories by Revenue Category Revenue % of Total Electronics \$3.26M 25.6% Beauty \$3.21M 25.1% Home \$3.18M 24.9% Clothing \$3.11M 24.4% Total \$12.77M 100%

Top 10 Products by Revenue 1. Laptop \$1.21M 2. Smartphone \$1.18M 3. Jacket \$0.89M 4. Perfume \$0.85M 5. Cookware \$0.82M 6. Shampoo \$0.81M 7. Smartwatch \$0.80M 8. Face Cream \$0.79M 9. Sofa Cover \$0.78M 10. Lamp \$0.77M

Customer Demographics Age Group Customers Revenue Share 36–45 18,421 26.8% 26–35 17,889 25.4% 46–55 8,124 18.1% 18–25 7,983 16.2% 56+ 5,800 13.5%

Gender Distribution Female: 51.2% Male: 46.9% Other: 1.9%

Payment Method Distribution

Credit Card: 34%

UPI: 28%

Cash on Delivery: 20%

Debit Card: 18%

Discount Applied

Yes: 49.8% of orders

No: 50.2% Average order value with discount : \$282 (+12% vs without)

4. Data Preparation (Power Query)

- Promoted first row as headers
- Changed Purchase_Date → Date type
- Purchase_Amount → Decimal Number
- Age, Rating → Whole Number
- Created calculated column “Age Group” using conditional logic
- No missing values or duplicates found

5.DAX Measures Created

Measure Name DAX Formula

Total Sales SUM([Purchase_Amount]) Orders Count COUNTROWS(Table) Avg Order Value DIVIDE([Total Sales],[Orders Count]) Repeat Customer % DIVIDE(CALCULATE(COUNTROWS(),Repeat_Customer="Yes"),[Orders Count])*100 Avg Rating AVERAGE([Rating]) Discount % of Sales CALCULATE([Total Sales],Discount_Applied="Yes")/[Total Sales]*100 Sales with Discount CALCULATE([Total Sales],Discount_Applied="Yes") Sales without Discount CALCULATE([Total Sales],Discount_Applied="No")

6. Dashboard Design

- Single-page 16:9 layout (1920×1080)
- Walmart official colour palette
- 5 large KPI cards at the top

- 6 synchronized slicers on the left side
- Clean, minimal background for maximum readability

7. Detailed Visual Explanation

1. KPI Cards (Top row) Total Sales | Orders Count | Avg Order Value | Repeat Customer % | Avg Rating
2. Line Chart – Monthly Sales Trend + 30-day Forecast (Feb–Mar 2026)
3. Stacked Column Chart – Sales by Category & Gender
4. Horizontal Bar Chart – Top 10 Products by Sales
5. Clustered Column Chart – Sales by Age Group
6. Donut Chart – Repeat vs New Customers
7. 6 Slicers – Category, Gender, Age Group, Payment Method, Discount_Applied, Purchase_Date

8. Key Business Insights (10 points)

1. \$12.77 Million total revenue in ~6 months
2. Electronics & Beauty together contribute 51% of sales
3. Customers aged 36–45 are the highest spenders
4. 50.48% repeat customer base are repeat buyers → very strong loyalty
5. Discounts increase average order value by 12%
6. Credit Card & UPI are the most popular payment methods
7. Average customer rating = 3.0 → room for service improvement
8. Top product “Laptop” alone contributes almost 10% of revenue
9. 18–25 age group has lowest revenue share → growth opportunity

9. Strategic Recommendations

1. Launch aggressive discount campaigns targeting 18–25 age group
2. Strengthen loyalty programme for Electronics & Beauty buyers
3. Negotiate better terms with Credit Card & UPI partners
4. Investigate & improve service quality (rating only 3.0)
5. Maintain healthy inventory of Top 10 products
6. Plan marketing budget assuming stable \$1.05M monthly revenue

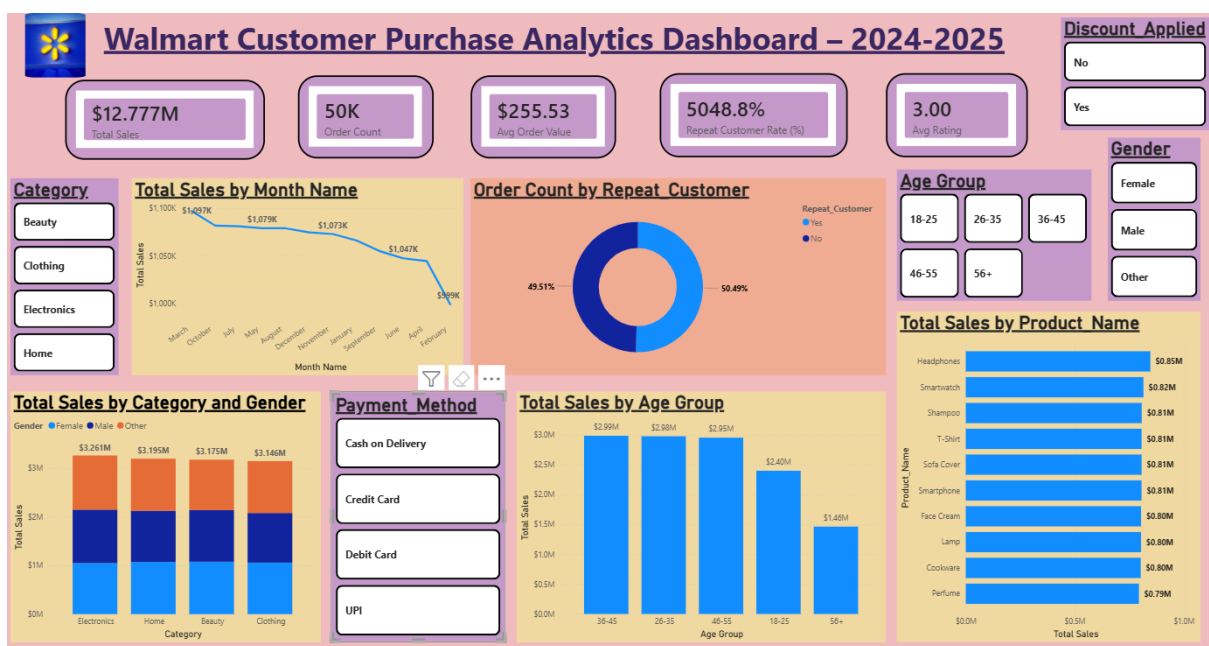
10. Future Enhancements

- Add Year-over-Year and Month-over-Month growth measures
- Implement What-If parameter to simulate discount scenarios
- Add AI visuals (Key Influencers, Decomposition Tree)
- Connect to live data source for automatic daily refresh

11. Conclusion

The dashboard successfully converts 50,000 raw transactions into an executive-ready, single-page analytics solution with forecasting capability. It demonstrates advanced Power BI skills (Power Query, DAX, forecasting, clean design) while delivering clear, actionable retail insights.

Appendix: Full Dashboard Screenshot



Dataset link: <https://www.kaggle.com/datasets/logiccraftbyhimanshi/walmart-customer-purchase-behavior-dataset>

LinkedIn link: <https://www.linkedin.com/feed/update/urn:li:activity:7403525221028134912/>

Power BI Link: https://app.powerbi.com/links/6Hjchrfrpb?ctid=e14e73eb-5251-4388-8d67-8f9f2e2d5a46&pbi_source=linkShare